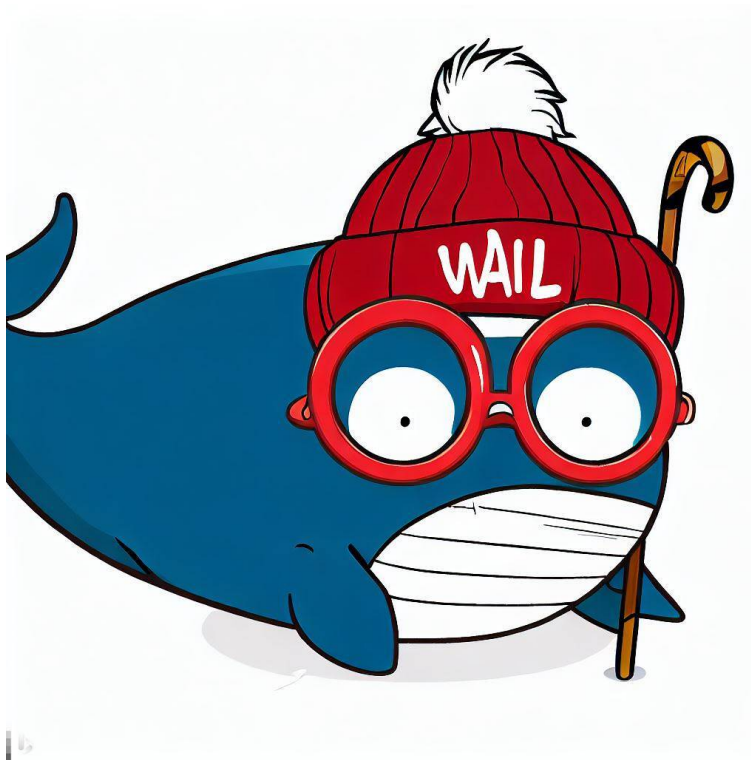


Where's Whaledo v2.0

User Manual



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A Overview

Where's Whaledo is a MATLAB-based toolkit for time-difference-of-arrival (TDOA) and direction-of-arrival (DOA) localization of odontocete echolocation clicks. The software is designed to generate fine-scale, three-dimensional tracks of diving, echolocating marine mammals and is optimized for use with small-aperture tetrahedral hydrophone arrays.

This manual is organized into several major sections and is intended to be read sequentially before beginning a project. Users may then revisit specific sections as needed during analysis. Questions, bug reports, or feature requests can be directed to Lauren Baggett (lbaggett@ucsd.edu) or submitted through the GitHub Issues page: <https://github.com/MarineBioAcousticsRC/Wheres-Whaledo>.

The original version of *Where's Whaledo* was developed by Dr. Eric Snyder. Version 2.0, developed by Lauren Baggett, introduces significant improvements in functionality and usability. Key updates include:

- resolution of known errors in Version 1.0 (see release notes for details)
- expanded support for tracking multiple species
- enhanced tools for track verification
- automatic saving of relevant parameters and metadata
- new 3D visualization and video export capabilities

Users of *Where's Whaledo* v1.0 are strongly encouraged to upgrade to v2.0 to take advantage of these improvements.

The following publications describe applications and development of *Where's Whaledo*:

- Baggett, L.M., Snyder, E.R., Solsona-Berga, A., et al. (2025). Long-term monitoring of *Ziphius cavirostris* behavior using 3D tracking from fixed hydrophone arrays off Southern California. *Scientific Reports*, 15, 40859. <https://doi.org/10.1038/s41598-025-24490-x>
- Snyder, E.R., Solsona-Berga, A., Baumann-Pickering, S., Frasier, K.E., Wiggins, S.M., et al. (2024). Where's Whaledo: A software toolkit for array localization of animal vocalizations. *PLOS Computational Biology*, 20(5), e1011456. <https://doi.org/10.1371/journal.pcbi.1011456>



B Getting Started

This chapter walks you through installing *Where's Whaledo* and creating your first project. By the end of this chapter, you will have:

- Installed and configured MATLAB
- Installed the *Where's Whaledo* software
- Created a working project



Figure B.1: *Where's Whaledo* home screen.



B.1 System Requirements

Where's Whaledo runs on any modern computer capable of running MATLAB and GitHub Desktop. No specialized hardware is required. For tracking large datasets, several gigabytes of available disk space are recommended for project files and output data.

Where's Whaledo has been tested on the following operating systems:

- Windows 11 or later
- macOS Sequoia (version 15) or later

Other operating systems may work but have not been formally tested.

B.2 Configuring MATLAB

Where's Whaledo requires a working installation of MATLAB and a valid MATLAB license. Many users have access to a MATLAB license through their university or institution. If you are unsure whether you have access, contact your institution's IT department or visit the MathWorks website (<https://www.mathworks.com/>) for licensing options.

MATLAB version requirements

Where's Whaledo requires MATLAB 2024a or newer. In addition to base MATLAB, the following toolboxes must be installed:

- Mapping Toolbox
- Parallel Computing Toolbox
- Signal Processing Toolbox
- Statistics and Machine Learning Toolbox

Follow the instructions below depending on whether you are installing MATLAB for the first time or checking an existing installation.

Installing MATLAB from scratch

If MATLAB isn't currently installed on your computer,

1. Download MATLAB version 2024a or newer from the MathWorks Downloads page (<https://www.mathworks.com/>)
2. Follow the on-screen instructions to install MATLAB and activate your license
3. During installation, select to install the required toolboxes listed above

Checking an existing installation

If MATLAB is already installed on your computer, verify that it meets the requirements to run *Where's Whaledo*:

1. Open MATLAB



2. In the command window, type `ver` and press Enter
3. Confirm that your MATLAB version is 2024a or newer, and that the required toolboxes are installed
4. If you are missing some toolboxes and need to install them, see the MathWorks installation guide (<https://www.mathworks.com/help/install/ug/add-to-existing-installation.html>).

B.3 Installing *Where's Whaledo*

Where's Whaledo is a code repository hosted on GitHub, and is in constant development and up-grading. To ensure that you receive any updates and bug fixes as they are released, it is strongly recommended that you access the repository through GitHub rather than downloading a static version of the code.

Step 1: Install GitHub Desktop

If GitHub Desktop is not already installed:

1. Download GitHub desktop from <https://desktop.github.com/>.
2. Run the installer and follow the on-screen instructions to log into your GitHub account.

Step 2: Clone the *Where's Whaledo* Repository

1. Open GitHub Desktop.
2. Select **File** → **Clone repository**.
3. Choose the **URL** tab.
4. Enter the repository address:
`https://github.com/MarineBioAcousticsRC/Wheres-Whaledo`
5. Choose a folder on your computer where you want to store the software.
6. Click **Clone**.

GitHub Desktop will link the *Where's Whaledo* software from GitHub to your computer.

Step 3: Add *Where's Whaledo* to the MATLAB Path

1. Open MATLAB.
2. In the MATLAB toolbar, select **Home** → **Set Path**.
3. Click **Add with Subfolders**.
4. Navigate to the folder where you cloned *Where's Whaledo* and select it.
5. Click **Save**, then **Close**.



Step 4: Open *Where's Whaledo*

To run *Where's Whaledo*, simply type `whereswhaledo` in the command window and press enter. The *Where's Whaledo* home screen (Figure [B.1](#)) will pop up if the software has been correctly installed.



B.4 Working with Projects

B.4.1 Projects Overview

In *Where's Whaledo*, a **project** is a self-contained workspace used to process data from an ensemble or deployment. Each project stores its analysis settings, metadata, and output files in a dedicated project folder. In most workflows, you will create one project per ensemble or deployment.

Projects are designed to keep analyses organized, reproducible, and easy to share. Configuration settings are saved automatically within the project, allowing you to return to a project later without reconfiguring parameters and to share the project folder with collaborators to reproduce the same workflow. Because each project maintains an independent configuration, you could work on multiple projects simultaneously without mixing data or settings.

When a project is created, *Where's Whaledo* generates a project folder at the location you specify. Project settings and metadata are stored in a nested folder named **config**, which contains the files required for the project to run correctly.

Important: Do not move or modify files inside the **config** folder. Changing or removing these files may cause errors or prevent the project from opening correctly.

B.4.2 Creating a project

To create a new project in *Where's Whaledo*:

1. Launch *Where's Whaledo* (see Section B.3).
2. Select **Create New Project** (Figure B.1).
3. Enter the required project information in the New Project wizard (Figure B.2).
4. Click **Save**.

A new project folder is created and initialized with your selected settings, and the project home screen opens automatically (Figure B.3). You can edit the project information at any time from this screen by selecting **Edit project metadata** in the upper-left corner.

B.4.3 Loading an existing project

To load an existing project, simply click **Load Existing Project** on the *Where's Whaledo* home screen (Figure B.1), then select the project folder in the popup file explorer window. When the project is successfully located, the project home screen opens automatically (Figure B.3).

B.4.4 Project Parameters

When creating a project, you will be asked to provide the following information:

Project Save Folder (folder location) The directory where the project folder will be created and stored. Choose a location with sufficient disk space for your analysis files.

Project Name (text field) A descriptive name for the deployment or dataset. This name is used to label the project folder and output files.



UserId (text field) Identification for the user initializing the project. It is best practice to input the same UserId that you would for Tethys (see <https://tethys.sdsu.edu/documentation/>).

Data Source (text field + dropdown menu) The acoustic data source associated with the project. For most users, this will be the ensemble name. Select whether the source represents an ensemble or a deployment.

An *ensemble* is a group of logically related instruments, such as two 4-channel arrays deployed simultaneously at the same site for localization (see <https://tethys.sdsu.edu/documentation/>).

Array Configuration (dropdown menus) Specifies how your instruments were oriented relative to the sea surface (horizontal or vertical) and whether your system contains one or two 4-channel arrays.

Software (automatic) Auto-filled software name, *Where'sWhaledo*

Software Version (automatic) Auto-filled software version information.

On-effort species (text field) Latin species names for the species you intend to track. Enter one name per line or provide a comma-separated list. Entries must use standard Latin binomial nomenclature.

Abstract (text field) A working abstract for this project, describing your general purpose and goals. This abstract can be edited as the project evolves.

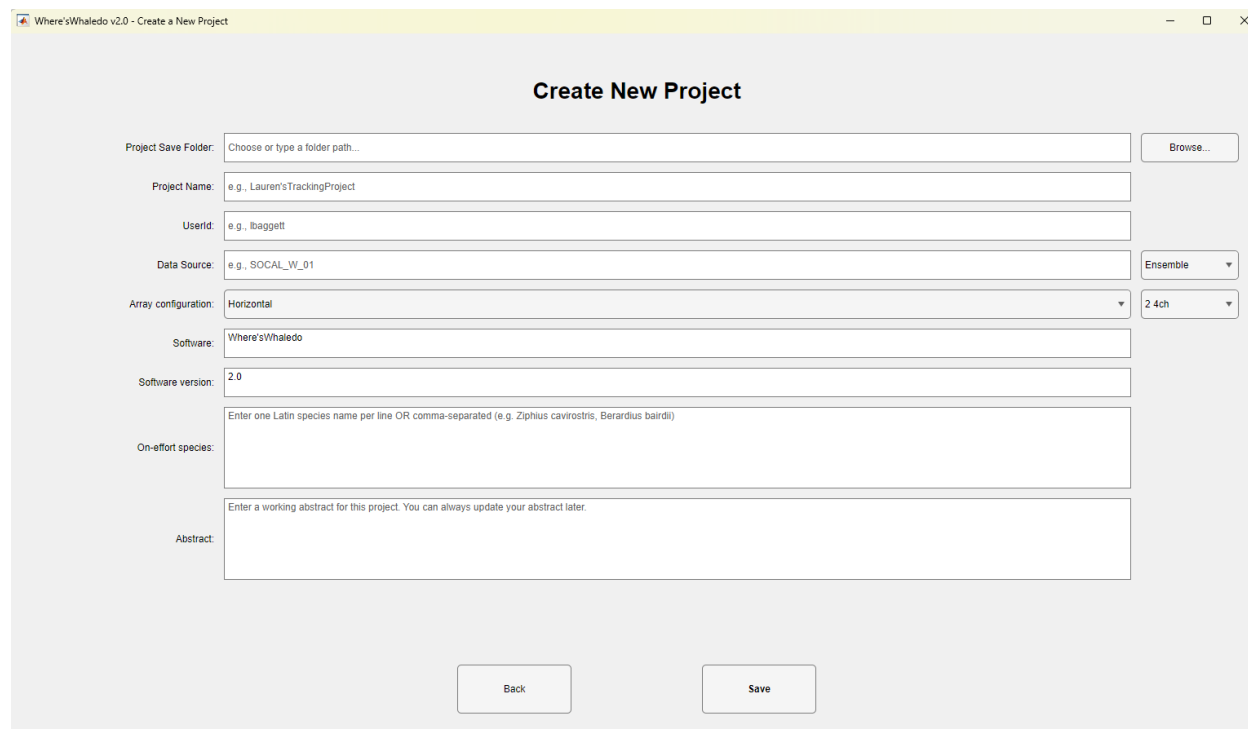


Figure B.2: *Where'sWhaledo* new project wizard screen.

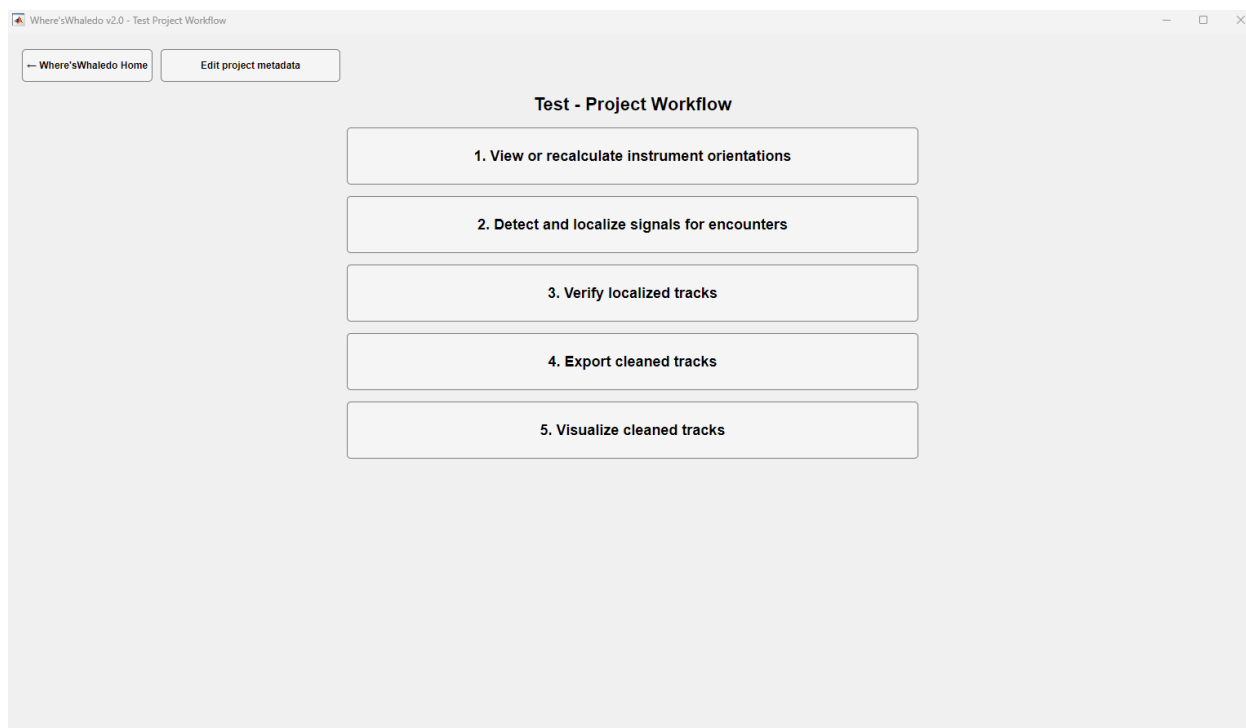


Figure B.3: *Where's Whaledo* project home screen.



1 Instrument Orientations

This chapter walks you through visualizing, calculating, and/or recalculating instrument orientations. By the end of this chapter, you will have:

- Visualized instrument orientations
- Calculated or recalculated instrument orientations if necessary



1.1 Overview of Instrument Orientations

In *Where's Whaledo*, **instrument orientations** describe the position of each receiver in a small-aperture array relative to a reference receiver (we use the first receiver in the array, often called channel 1). These relative positions define the geometry of the array used during whale localization calculations.

Accurate instrument orientations are essential for successful localization of diving whales. If these positions are incorrect, localization results will be inaccurate and unstable.

Orientations are calculated using broadband engine noise from a ship, often the vessel that deployed or recovered the instrument during the localization period. Time differences of arrival (TDOAs) between receivers are combined with the known ship positions and solved as a system of equations to estimate the relative hydrophone positions within the array. A detailed description of the underlying math is provided in Snyder et al., 2024.

Instrument orientations and metadata are stored in a nested folder within the project named `instrument_orientations`. Changing the names of the files within this folder or moving them to a different folder will cause errors in future calculations.

1.2 Visualizing Existing Instrument Orientations

For many ensembles, instrument orientations have already been calculated (Baggett et al., 2025) and are included with *Where's Whaledo* v2.0. If your ensemble is supported, the orientations are loaded automatically when you open your project.

A list of ensembles with pre-calculated orientations is available in the *Where's Whaledo* GitHub repository: https://github.com/MarineBioAcousticsRC/Wheres-Whaledo/tree/main/ww/instrument_orientation/precalculated_ensembles

Even when orientations are pre-calculated, you should visually inspect them as a sanity check. To view instrument orientations, select **1. View or recalculate instrument orientations** on the project home screen (Figure B.3). The orientation visualization window opens automatically (Figure 1.1).

The visualization screen contains three main panels for each instrument (Figure 1.1):

- The top panel displays the orientation file name and associated metadata, including array configuration, sound speed, and instrument coordinates.
- The middle panel shows a rotatable 3D view of the receiver positions within the array. Rotate the figure to confirm that it matches the expected tetrahedral geometry.
- The bottom panel displays the distances (in meters) between each pair of hydrophones. Compare these values to the measurements recorded before deployment and verify that they are consistent with the expected spacing.

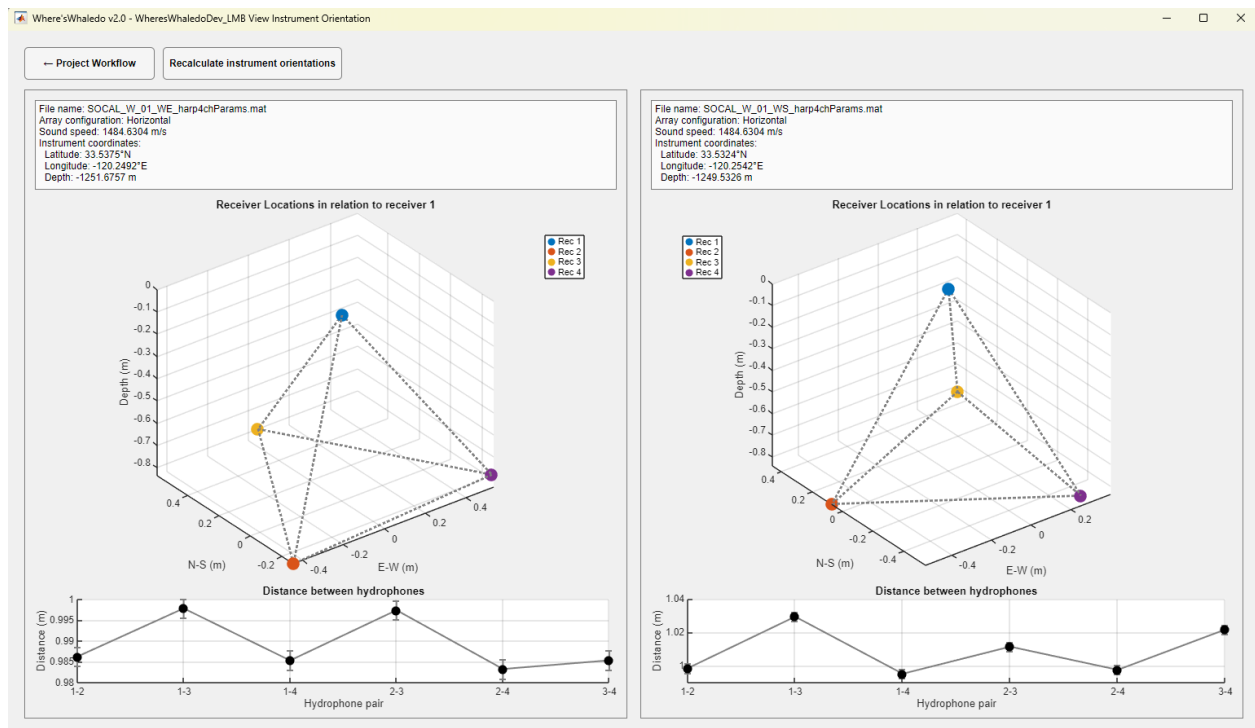


Figure 1.1: *Where's Whaledo* instrument orientations visualization screen.



1.3 Calculating or Recalculating Instrument Orientations

For newer deployments, instrument orientations have not yet been calculated and you will need to calculate them yourself. In this case, click the button on the project home screen labeled **1. Calculate instrument orientations** (Figure B.3) to open the instrument orientation calculation wizard (Figure 1.2). If you'd like to recalculate instrument orientations, click **Recalculate instrument orientations** from the top left of the instrument orientations visualization screen (Figure 1.1).

From here, the steps to calculate instrument orientations are as follows:

1. Enter the required information into the calculate instrument orientation onscreen wizard. For more information about each setting, see section 1.3.1.
2. Click **Calculate** and wait for the progress bar to progress. This should take several minutes.
3. Once the TDOAs have been calculated, the screen will update. Click to open the interface to inspect your TDOAs (Figure 1.3).
4. Following the instructions, inspect your TDOAs. The interface displays time vs TDOA for each hydrophone pair. In red are the expected TDOAs, from the GPS information. In blue are the observed TDOAs from the acoustic data. Brush any outlier points until the observed TDOAs match well with the expected TDOAs (Figure 1.4).
5. Once you're finished brushing, click **Save & Continue** (Figure 1.3). The screen will automatically redirect you to the instrument orientation visualization page (Figure 1.1).
6. Inspect your calculated orientation as described in section 1.2.

Where's Whaledo v2.0 - WheresWhaledoDev_LMB Calculate Instrument Orientation

← Project Workflow

Calculate instrument orientation

Method: ☒ from TTGPS ☐ from AIS

Path to xwav files: Choose or type a folder path... Browse...

Path to TTGPS files: Choose or type a file path... Browse...

Instrument Name: e.g., SOCAL_W_01_WE

Instrument location: Lat (°N) Lon (°E) Depth (m)

Localization date: yyyyMMdd Sound speed (m/s): 1490

Load segment (s): 30 Xcov window (s): 1

Bandpass filter edges (Hz): 100 10000

Back Calculate

Figure 1.2: *Where's Whaledo* instrument orientation calculation wizard screen.

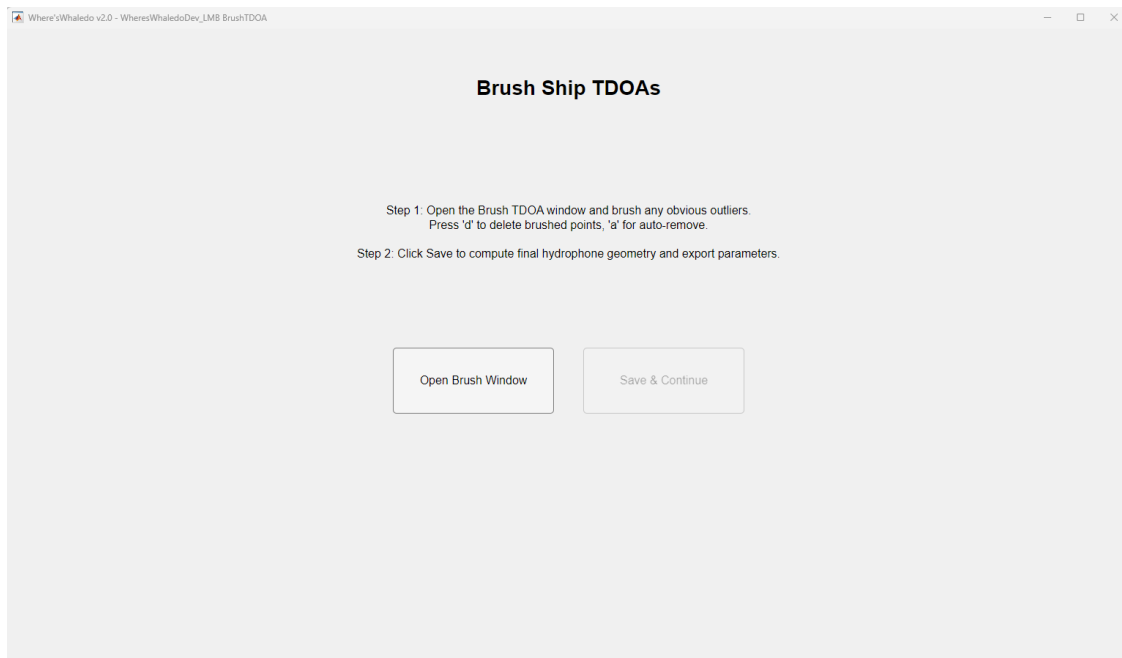


Figure 1.3: *Where's Whaledo* screen to launch ship TDOAs.

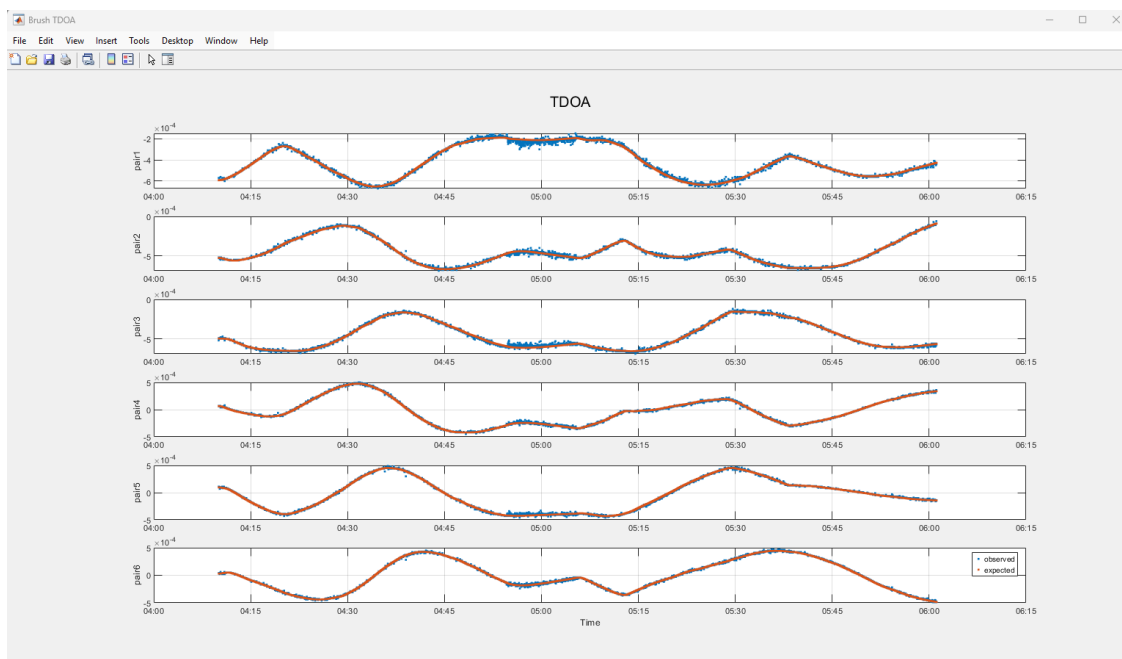


Figure 1.4: *Where's Whaledo* screen comparing expected TDOAs from ship GPS information to actual TDOAs from acoustic data.



1.3.1 Instrument Orientation Parameters

Method (selected button) You can run this calculation using either GPS locations collected on the ship during instrument localization (TTGPS files), or from AIS data. It is recommended to use TTGPS locations rather than AIS locations when possible to ensure higher precision.

Path to xwav files (folder location) File path to xwav file disk. *Where's Whaledo* will search recursively through the disk to find xwav files.

Path to ship location files (folder location) File path to either TTGPS information or AIS information. Expected format for AIS data is a .csv file with columns "LAT," "LON," and "BaseDateTim" with time in the format yyyy-MM-ddTHH:mm:ss.

Instrument Name (text input) Name of the instrument you're calculating orientations for. Note that if you are working with an ensemble containing two 4-channels, you'll need to run this instrument orientation calculation for each instrument.

Instrument Location (text inputs) Input the latitude ($^{\circ}$ N), longitude ($^{\circ}$ E), and depth (m) of the instrument. MBARC instruments are localized on deployment and/or recovery to find the array landing positions. **Tip:** Inspect the ship track and localization residuals before accepting the estimated landing position. Large residuals may indicate that the position should be recalculated.

Localization date (text input) Date (UTC time) during which the instrument was localized, or the date for which you have AIS data, in the format yyyyMMdd.

Sound speed (text input) Sound speed (m/s) calculated from either in-situ CTD measurements (preferred) or modeled temperature and salinity data.

Load segment (text input) Specify the duration of audio data (in seconds) loaded at one time from the xwav files during processing.

Xcov window (text input) Specify the duration of audio data (in seconds) used for cross-correlation when calculating ship TDOAs.

Bandpass filter edges (text inputs) Frequency edges (Hz) for the bandpass filter. **Tip:** The ship should be the highest amplitude signal within this frequency band.



1.4 Instrument Orientations Troubleshooting

Below are some common problems alongside their solutions.

Problem: I expected that my instrument orientations would be pre-calculated, but none appeared on my project home screen.

Solution: Verify that the ensemble name in your project matches the spelling used in the repository. Verify that the ensemble exists in the repository (https://github.com/MarineBioAcousticsRC/Wheres-Whaledo/tree/main/ww/instrument_orientation/precalculated_ensembles).

Problem: I don't see any track at all in my calculated TDOAs, just a bunch of randomly scattered points.

Solution: Double check that your timestamps are aligned correctly. Are your GPS inputs in UTC time? Did you input the correct date for localization? Open the LTSA and find where the localization happened to be extra sure.

Problem: My actual TDOAs are orders of magnitude larger than my expected TDOAs, and my calculated orientations are way too big.

Solution: Likely you are cross-correlating a signal that isn't your ship. Is the ship the highest amplitude signal in your frequency band? Check in the LTSA to find a frequency band that could work better and rerun with adjusted parameters.

Problem: My calculated TDOAs are on the same order of magnitude as my expected TDOAs and my calculated orientation looks almost right, but they don't quite match the expected values.

Solution: Likely your instrument landing position or calculated sound speed is off. Revisit the calculations for both of these parameters, and then run again with updated values.



2 Detection & Localization

This chapter walks you through running the detection and localization steps. By the end of this chapter, you will have:

- Detected and localized signals from encounters of interest
- Integrated species labels from a machine learning workflow, if desired



2.1 Overview of Detection & Localization

Accurate localization depends on first identifying individual echolocation clicks, since these detections provide the timing information used to estimate source direction. To track diving odontocetes, *Where's Whaledo* first detects candidate echolocation clicks and then computes the direction of arrival (DOA) of each detected signal. In most deployments, this process is performed for multiple 4-channel instruments. The resulting DOAs are then cross-fixed to estimate a three-dimensional (3D) position for each detected whale. If only a single 4-channel instrument is available, tracking is still possible, but localization is limited to two dimensions (azimuth and elevation) rather than full 3D position.

Below is a brief overview of the detection and localization steps. For more details about the algorithm, including equations, see Snyder et al., 2024.

Detection The click detection stage isolates impulsive, high-frequency signals from background noise and identifies them as candidate echolocation clicks. The detector operates in the time domain, passing the waveform through a fourth-order, zero-phase elliptical high-pass or bandpass filter to emphasize click energy and suppress out-of-band noise. After filtering, peaks that exceed a fixed amplitude threshold are marked as potential detections. A short segment of waveform surrounding each candidate click is retained for comparison across receivers during the localization step.

Localization *Where's Whaledo* uses time-difference-of-arrival (TDOA) localization to estimate the direction of a sound source from arrival times recorded on multiple receivers. Differences in arrival time between receiver pairs constrain the possible source location. The software is designed for use with **small-aperture** arrays, where receiver spacing is much smaller than the distance to the sound source. Under this condition, the incoming signal can be approximated as a plane wave traveling across the array. A click reaches one receiver slightly before the others, and these small timing differences reveal the angle at which the wave passes through the instrument.

Each receiver pair provides a geometric constraint on source direction. When constraints from all pairs are combined, they define a single line pointing outward from the array toward the whale. This line is the direction of arrival (DOA). In the software, the DOA is represented using two angles: azimuth and elevation. Azimuth describes the horizontal direction (measured counterclockwise, with 0° pointing east and 90° pointing north), and elevation describes the vertical direction (0° downward, 90° horizontal, and 180° upward toward the sea surface). In summary, detected clicks are converted into timing differences between receivers, which define directions of arrival that can then be combined to estimate whale positions.



2.2 Running the Detector & Localizing Detections

To run the detection and localization step for your ensemble or deployment, follow these steps:

1. Identify the start and end times of encounters of interest during your on-effort time period. Ideally, these encounters are no longer than an hour or so. For particularly long encounters, it might be beneficial to split them for detection-localization and track cleaning, then stitch them back together in post-processing.
2. On your project home screen (Figure B.3), click **2. Detect and localize signals for encounters** to open the detection-localization onscreen wizard.
3. Enter the required information into the detection-localization onscreen wizard (Figure 2.1). For more information about each setting, see section 2.2.1.
4. Click **Run detector** and wait for the progress bar to progress. Depending on how many encounters you are running, this step could take several minutes or several hours.
5. Once the detector has finished running, you will be redirected to the project home screen.
6. On your computer, navigate to your output folder to verify that detection-localization files were saved.

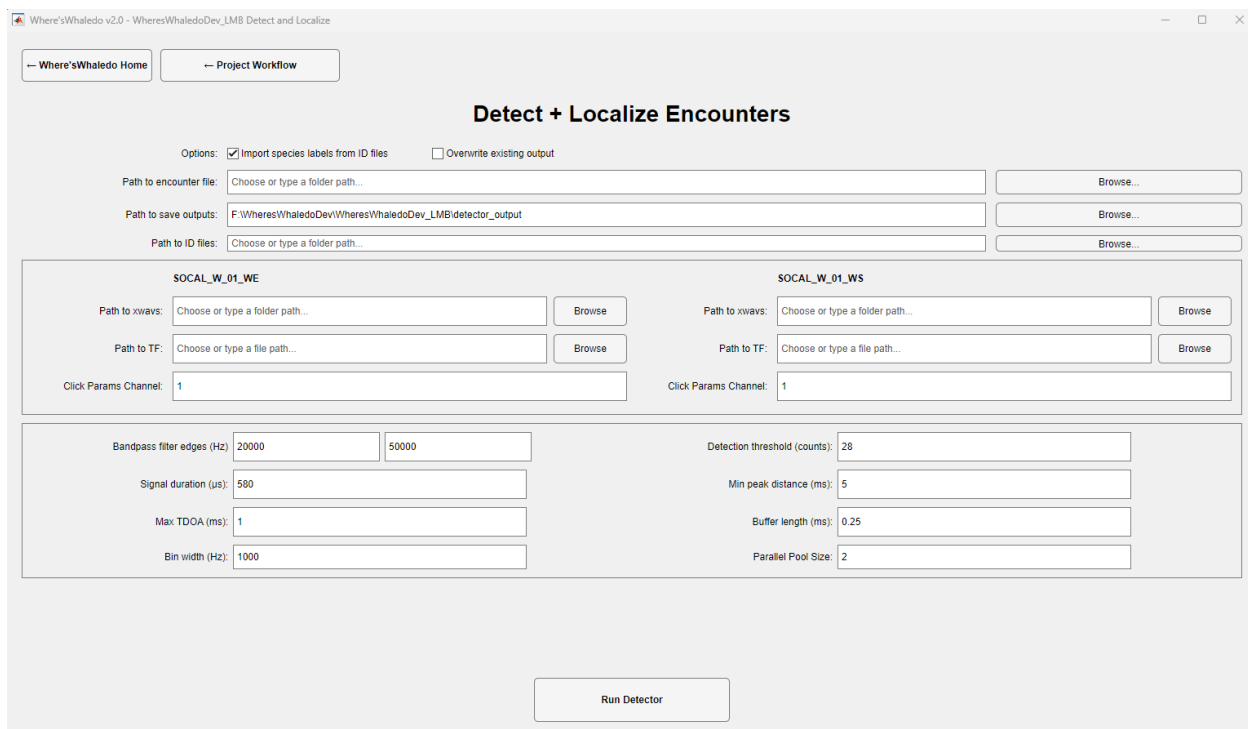


Figure 2.1: *Where's Whaledo* detection-localization wizard screen.



2.2.1 Detection & Localization Parameters

Options (select checkboxes) Optional checkbox settings that control how detection is performed. Check **Import species labels from ID files** to include existing species labels in the output. Check **Overwrite existing output** to rerun the detector from the beginning. If this option is unchecked, the detector resumes from where it previously stopped and will not rerun encounters that have already been processed.

Path to encounter file (file location) File path to the encounter file containing times of interest. See the folder [ww/helper_scripts](#) for help making this encounter file.

Path to save outputs (folder location) Folder where output files will be saved. By default, this is a folder within your project directory, but it can be changed if needed.

Path to ID files (folder location) Optional folder containing species ID files. If you have ID files for multiple 4-channels in your ensemble (rare), make sure they're all in this same folder.

Path to xwavs (folder location) For each 4-channel instrument in your ensemble, file path to the xwav disk. *Where's Whaledo* can search recursively within this location to find xwav files.

Path to TF (file location) For each 4-channel instrument in your ensemble, file path to the corresponding transfer function (.tf) file. **Tip:** Make sure that your TF matches your selected channel.

Channel (text input) Specifies which channel of the 4-channel array is used for signal detection and click parameter calculations.

Bandpass filter edges (text inputs) Frequency edges (Hz) of the bandpass filter used during detection. **Tip:** Choose edges that contain most of the echolocation click energy for your species of interest.

Signal duration (text inputs) Expected signal duration (μ s) for clicks of interest. **Tip:** Consult the literature for typical click durations for your species.

Max TDOA (text input) Maximum allowed time difference of arrival (ms) used when cross-correlating signals across receivers.

Detection threshold (text input) Amplitude threshold (in counts) used to identify candidate clicks. **Tip:** Use a threshold low enough to capture all clicks within a track.

Min peak distance (text input) Minimum allowable time separation (ms) between successive detections.

Buffer length (text input) Time buffer (ms) added on either side of each detection when cross-correlating across channels.

Bin width (text input) Frequency bin width (Hz) used for FFT calculations.

Parallel Pool Size (text input) Number of parallel workers used during processing. **Tip:** Check how many CPU cores your computer has before increasing this value.



2.3 Detection & Localization Troubleshooting

Below are some common problems alongside their solutions.



3 Track Verification

This chapter walks you through verifying your tracks and finding final 3D positions. By the end of this chapter, you will have:

- Verified tracks within encounters of interest
- Assigned whale numbers and species labels to each verified track in an encounter



3.1 Overview of track verification

The previous detection step is intentionally designed to find as many potential echolocation clicks as possible, and subsequently detects a lot of signals that you might not be interested in. You may have detected signals from a variety of sources, including:

- Clicks produced by species outside your study focus
- Anthropogenic signals (e.g. echosounders)
- Detections from instruments on the mooring (e.g. an ADCP or WBAT echosounder)

In addition, multiple individuals are often present at the same time during an encounter. To effectively track individual whales, it is important to review the automatically generated tracks. After detection, you should remove detections not relevant to your study, separate signals from different individuals and assign each a unique number, and assign species labels to tracked whales. These steps are executed in the track verification interface of *Where's Whaledo*.

Track verification happens both in 2D (azimuth-elevation) space, and in 3D space. Each 4-channel array estimates the direction-of-arrival (DOA) of a signal. Each DOA can be thought of as a unit vector, or a line, extending outward from the array in the direction that the signal came from. To estimate the source location, the software:

1. Takes the DOAs labeled with the same whale number that are closest together in time from each array
2. Estimates the point along the DOA lines from each array where they are closest to intersecting
3. Solves the system of equations relating the source location to the directions of arrival to produce a final estimate for source location

For more information on how this algorithm works, including equations, see Snyder et al., 2024.

3.2 Running the track verification interface

To run the track verification steps for your detected and localized encounters, follow these steps:

1. From your project's home screen (Figure B.3), select **3. Verify localized tracks**.
2. Select a folder location for your detection outputs, and then select an encounter file to verify. Once you've clicked on the file you'd like to verify, click **Display** to open the *BrushDOA* interface.
3. The interface that opens is comprised of multiple screens. First, the *Where's Whaledo* screen will display with new inputs for your selected encounter (Figure 3.2). For a description of each section of this screen, see section 3.3. Next, you'll see the *BrushDOA* interface with associated additional windows. For a description of each section of this interface, see section 3.3.
4. Clean the encounter by removing detections not produced by your species of interest, and then assign whale number labels and species labels for each track in this encounter.
5. When you're finished cleaning, choose a save location and then click **Save labels for this encounter**.

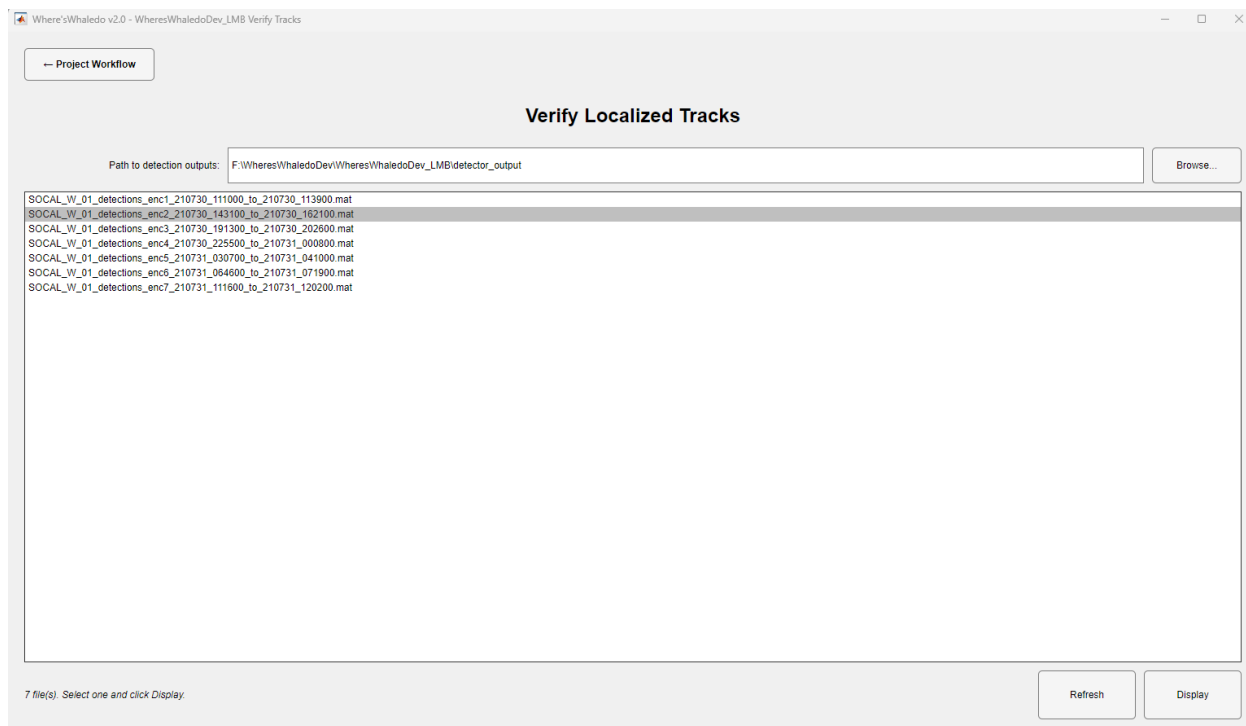


Figure 3.1: *Where's Whaledo* track verification encounter selection screen.

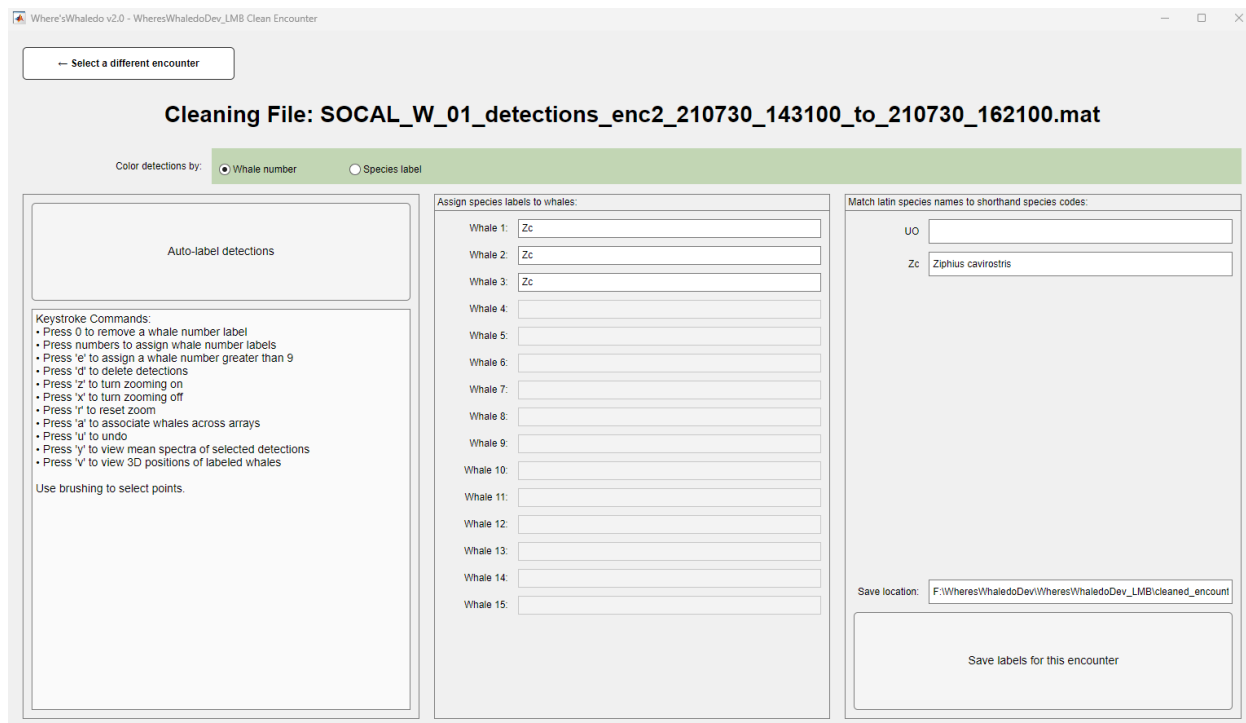


Figure 3.2: *Where's Whaledo* screen with inputs for your selected encounter.
For a description of each section of this screen, see section 3.3.

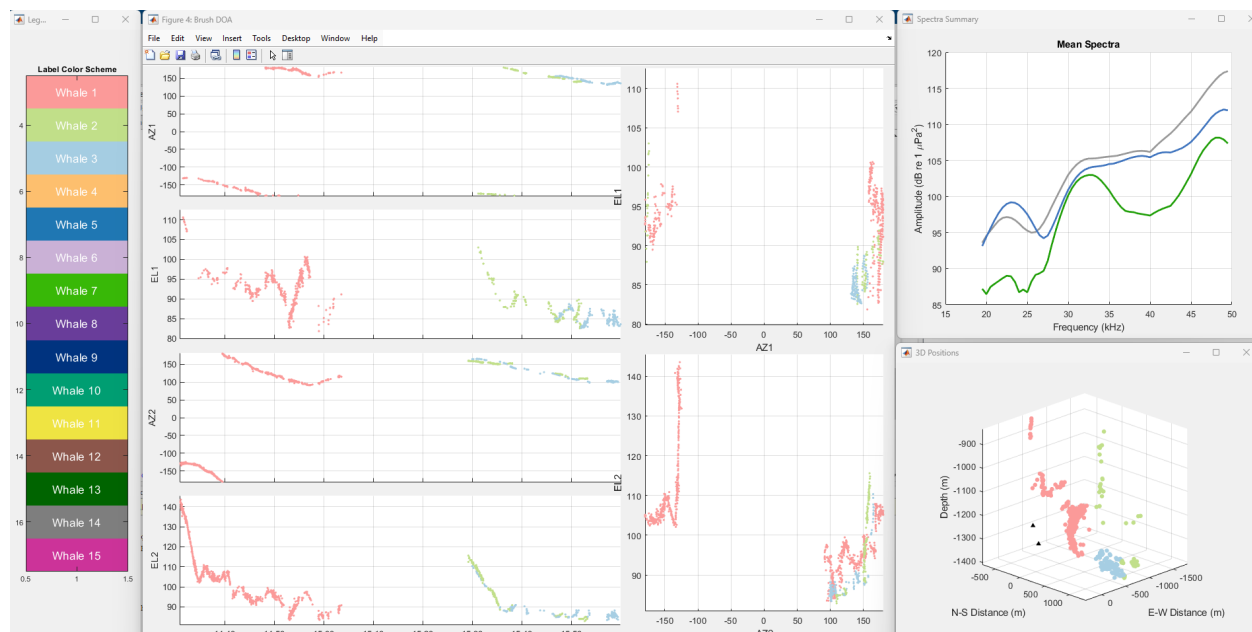


Figure 3.3: *BrushDOA* interface displaying a selected encounter, colored by whale number. For a description of each section of this screen, see section 3.3.

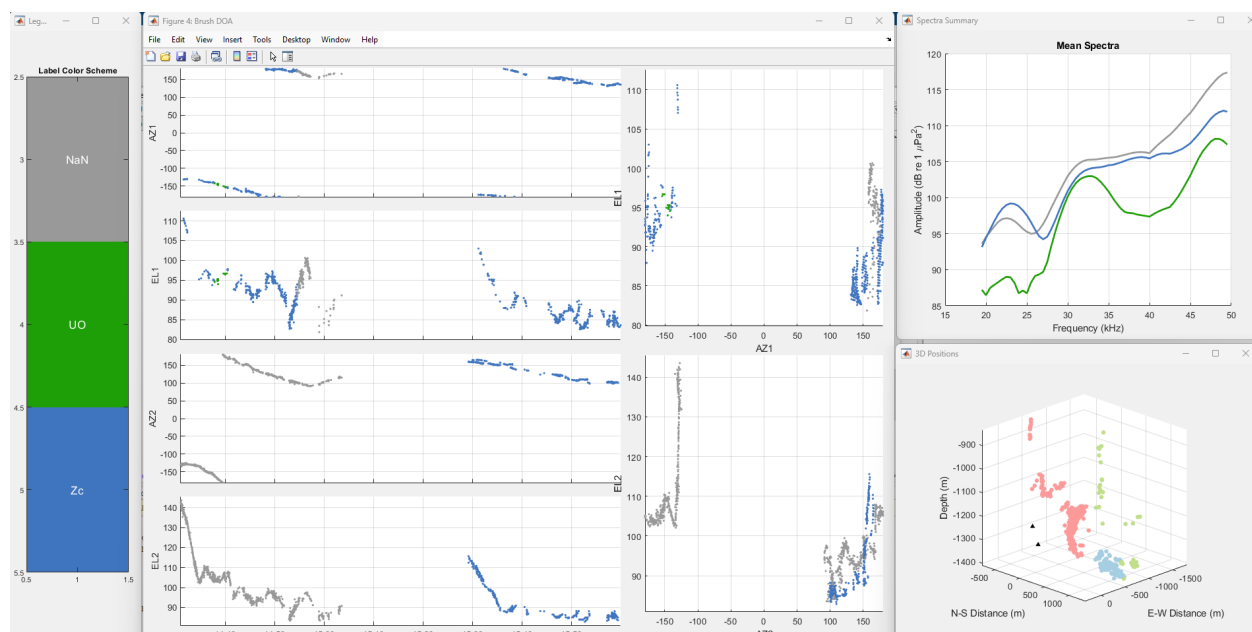


Figure 3.4: *Where'sWhaledo* screen with inputs for your selected encounter, colored by species label. For a description of each section of this screen, see section 3.3.



3.3 Track cleaning keystrokes and inputs

This section will outline all of the keystroke commands and inputs for track cleaning.

First, a description of each screen:

- *Where's Whaledo* verification screen for a selected encounter (Figure 3.2)
- Label color legend (Figure 3.3; 3.4 left). This will display either whale number colors or species label colors, depending on which input you select on the *Where's Whaledo* verification screen.
- *BrushDOA* interface (Figure 3.3; 3.4 middle). For each array in your ensemble, this interface displays azimuth and elevation values over time.
- Mean spectra for your detections (Figure 3.3; 3.4 top right), colored by species label.
- 3D positions of your labeled tracks (Figure 3.3; 3.4 bottom right), colored by whale number label.

Now, a description of the *Where's Whaledo* verification screen for a selected encounter (Figure 3.2).

Color detections by (radio button input): Select either whale number or species label. The color legend and *BrushDOA* interface (Figure 3.3; 3.4) will update accordingly based on your selection. If you didn't include species labels from ID files in your detection step, the species colors will all initially be NaN.

Auto-label detections (Available in a future release): Click this button to run a DBSCAN clustering algorithm that will automatically assign labels to individual whales within the encounter.

***BrushDOA* Keystroke commands:** When you're clicked into the *BrushDOA* interface screen, use these keystroke commands to clean and label an encounter:

- Press '0' to remove a whale number label.
- Press number keys 1-9 to assign whale number labels.
- Press 'e' to assign a whale numbers 10-15. **Tip:** if you have more than 15 individuals in an encounter, you should probably go back and break your encounter into smaller time durations. It really gets tough to reliably label more than 15 individuals, and you'll have better performance with a greater number of shorter encounters.
- Press 'd' to delete detections. *Tip:* Delete detections which are not clearly associated with a track, or those which are not produced by your species of interest.
- Press 'z' to turn zooming on. **Tip:** Use this feature to zoom in on tracks which appear close together in azimuth or elevation to better differentiate individuals.
- Press 'x' to turn zooming off and re-enter brushing mode.
- Press 'r' to reset zoom.
- Press 'a' to associate whales across arrays. This uses click-train-correlation (see Snyder et al., 2024) to match whale labels from one array to another. **Tip:** Begin by labeling whales on whichever array it is easier to differentiate individuals on. Completely clean this array, removing all erroneous detections and assigning whale number labels to each individual.



After this, associate between arrays and completely clean the second array as you did the first, assigning number labels based on the output of the association.

- Press ‘u’ to undo. This remembers only your previous keystroke command.
- Press ‘y’ to view mean spectra of selected detections. **Tip:** Use this feature to assign species labels to your delineated individual whales, and to determine which signals were/were not produced by your species of interest.
- Press ‘v’ to view 3D positions of labeled individual whales. **Tip:** Always check the 3D positions before you finish cleaning and save. Make sure that the 3D positions are reasonable and that the individual labels that you’ve assigned form logical tracks in 3D space as well as in 2D azimuth-elevation space.

Auto-label detections (Available in a future release): Click this button to run a DBSCAN clustering algorithm that will automatically assign labels to individual whales within the encounter.

3D Positions Keystroke commands: When you’re clicked into the 3D track viewer screen, use these keystroke commands to clean and label an encounter:

- Press ‘t’ to enter or exit rotation mode. While in rotation mode, you can turn the 3D figure to view it from different angles.
- Press ‘s’ to show selected points in the 3D figure on the *BrushDOA* screen.

Assign species labels to whales: For each whale that you delineate in the *BrushDOA* interface, you must assign a matching species label. Inputs for each whale number will become editable once you have labeled an individual whale with that number. If you added species labels from ID files in your detection step, then these species labels may auto-populate using the labeling code defined within your ID files.

Match latin species names to shorthand species codes: In most cases, the species labels contained in your ID file will be some shorthand species code rather than a full latin species name. For example, “Zc” is the shorthand species code commonly used for goose-beaked whale, *Ziphius cavirostris* (Figure 3.2). You must fill in this section to match your shorthand codes back to a real latin species name. This is the species label that will be included in your saved tracking files.

Save location: This field will auto-populate to a folder within your project called “cleaned_encounters.” It will create an individual folder within this one for this specific encounter. It is recommended to save files from each encounter within the same subfolder for organization moving forward.

Save labels for this encounter: Click this button to save both the whale number labels and species labels that you’ve assigned for this encounter. Multiple files will be saved when you click this button:

1. A cleaned DET file (azimuth-elevation information for detections on each array), containing only the detections which you’ve retained along with their whale number and species labels.
2. A whale struct file (3D position information for whales successfully detected on both arrays).
3. A MATLAB 3D figure showing the localized positions of each whale in the encounter.
4. A MATLAB 3D figure showing the smoothed 3D positions of each whale in the encounter. Whale positions are smoothed using a kalman filtering approach (see Baggett et al., 2025).



3.4 Track Verification Troubleshooting

Below are some common problems alongside their solutions.

Problem: There are no clear tracks in my file for a given encounter.

Solution: Either there were not enough signals detected to generate a track, or there were so many signals detected that delineating individual tracks is impossible. Likely, this encounter will be discarded. If you want to track groups of individuals rather than one individual at a time, you will need to make adjustments to the algorithm.

Problem: My tracks look fine in azimuth-elevation space, but when I look at the cross-fixed positions in 3D they're not logical (either much too far away, or all positions are collapsed into the plane crossing through both 4-channels).

Solution: This almost always happens when you've misassociated detections between the arrays. If it's just a few points within the track that have this issue, double check that you don't have a stray, mislabeled color somewhere within the track. If it's the entire track, you'll need to rerun the association step. Double check labels on whichever array has clearer tracks, and then unlabel the other array. Rerun association and carefully re-assign species labels based on the output. If you're having this problem for all of your encounters, particularly if the positions are collapsing into one plane, then your instrument orientations may also have an error (more likely if you calculated them yourself, since those which pre-populate have been verified).



4 Exporting Tracks

This chapter walks you through exporting your tracks from `.mat` files to a different format. Currently supported formats are `.nc` (optimized for upload to a DRYAD repository) and `.xml` (optimized for upload to Tethys).

By the end of this chapter, you will have:

- Exported your tracks to either `.nc` or `.xml` format



4.1 Overview of track exporting

Where's Whaledo operates within MATLAB, and its native data outputs are stored as `.mat` files. While this format is well-suited for analysis within MATLAB, users may need to export data for use in external software or for archival in databases such as DRYAD or Tethys. The currently supported export formats are `.nc` and `.xml`. Because requirements may vary by dataset, users should review exported files to ensure that all necessary metadata are included for their intended use.

4.2 Running the track export wizard

To export your detected, localized, and verified encounters, follow these steps:

1. From your project's home screen (Figure B.3), select **4. Export cleaned tracks** to open the track export wizard.
2. Enter the required information into the track export onscreen wizard. Depending on whether you are exporting to `.nc` (Figure 4.1) or `.xml` (Figure 4.2) the inputs are slightly different.
3. Click **Export** and wait. Depending on how many encounters you are exporting, this step could take several minutes.
4. Once the detector has finished running, the printed status will say "export complete"
5. On your computer, navigate to your output folder to verify that exported files were saved.

Figure 4.1: *Where's Whaledo* track export wizard screen for `.nc` files.



Where'sWhaledo v2.0 - Test Export Cleaned Tracks

Project Workflow

Export cleaned tracks

Export type: ☐ to .nc ☒ to .xml

Input folder: Choose or type a folder path... Browse...

Output folder: Choose or type a folder path... Browse...

Encounter file: Choose or type path to encounter spreadsheet... Browse...

Userid: lbaggett

Tethys path: File path to Tethys source code (install at <https://tethys.sdsu.edu/install/>)

Server: breach.ucsd.edu

Port: 9779

Status:

Export

Figure 4.2: *Where'sWhaledo* track export wizard screen for .xml files.



5 Visualizing Cleaned Tracks

This chapter walks you through using *Where's Whaledo v2.0* to generate figures from your cleaned tracks. Bathymetry data is extracted from GMRT (<https://www.gmrt.org/>) and colormaps from the `cmocean` package (<https://matplotlib.org/cmocean/>). These figures are intended as a starting point for data exploration and quality control. In most cases, additional customization and figure development will be required for more advanced analyses, presentations, and publication-quality outputs.

By the end of this chapter, you will have:

- Made spaghetti and density plots for all cleaned encounters in your project
- Made spaghetti plots and .mp4 movies for specific cleaned encounters of interest in your project.



5.1 Generating figures for all encounters in your deployment

To generate figures which aggregate tracks across encounters, follow these steps:

1. From your project's home screen (Figure B.3), select **5. Visualize cleaned tracks** to open the track visualization wizard.
2. Enter the required information into the track visualization onscreen wizard (Figure 5.2). Depending on whether you are generating spaghetti or density plots (Figure 5.1) the inputs are slightly different (see section 5.1.1).
3. Click **Plot** and wait. Depending on how many encounters you are plotting, this step could take several minutes.
4. Once the figure is generated, adjust dimensions and save accordingly. The figures are NOT auto-saved.

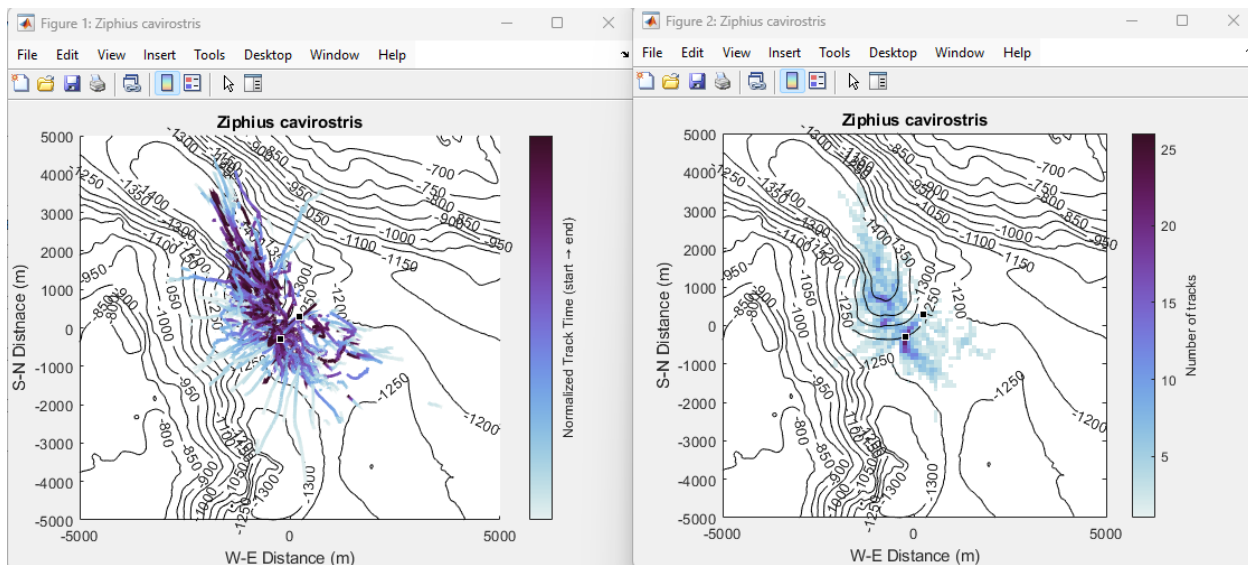


Figure 5.1: Examples of spaghetti (left) and density (right) figures.



Figure 5.2: *Where's Whaledo* track visualization wizard screen.

5.1.1 Visualize tracks across a deployment parameters

Plot type (select checkboxes) Checkboxes to generate each type of plot. Spaghetti plots show all tracks on the same figure, colored by normalized track time (a gradient from the start to the end of the track). Track density heatmaps show the number of tracks that pass through a defined grid square.

Path to cleaned encounters (folder location) File path to the folder containing cleaned encounters. All encounters in this folder will be shown on the figures.

Species plotting (dropdown selection) Choose whether to create separate figures per species or to combine all species on the same figure.

Select colormap (dropdown selection) Select a colormap for your figure(s).

Bathymetry contour step (text input) Input intervals (m) to draw contour lines at. The smaller the step the more contour lines you will have and the longer it will take to generate a figure.

Density bin size (text input) If you're generating a density heatmap, define the bin size (m) over which to aggregate tracks.

5.2 Generating figures for specific encounters of interest

To generate figures which show only tracks in specific encounters, follow these steps:



1. From your project's home screen (Figure B.3), select **5. Visualize cleaned tracks** to open the track visualization wizard.
2. Enter the required information into the track visualization onscreen wizard (Figure 5.2 or 5.3). Depending on whether you are generating a spaghetti plot or a movie the inputs are slightly different.
3. Click **Plot** and wait. Depending on how many encounters you are plotting, this step could take several minutes.
4. Once the figure is generated, adjust dimensions and save accordingly. The spaghetti figure is NOT auto-saved, but the movie will save in your specified output location.

Where's Whaledo v2.0 - Test Visualize Cleaned Tracks

← Project Workflow

Visualize Cleaned Tracks

Visualize tracks across a deployment

Plot type: ☒ Spaghetti plot ☒ Track density heatmap

Path to cleaned encounters:

Species plotting:

Select colormap:

Bathymetry contour step (m):

Density bin size (m):

Status:

Visualize tracks within individual encounters

Plot type: ☐ Spaghetti plot ☒ Movie (.mp4)

Path to selected encounter:

Color by:

Azimuth viewing angles:

Elevation viewing angles:

Save path:

Status:

Figure 5.3: *Where's Whaledo* track visualization wizard screen when generating a movie.



5.2.1 Visualize tracks within individual encounters parameters

Plot type (radio buttons) Select whether you'd like to generate a spaghetti plot or a .mp4 movie for this encounter.

Path to selected encounter (folder location) File path to the folder containing the cleaned whale struct for your selected encounter.

Color by (dropdown selection) Choose whether to color by individual whale number, by species, or by normalized time (for spaghetti plot only).

Bathymetry contour step (text input) Input intervals (m) to draw contour lines at. The smaller the step the more contour lines you will have and the longer it will take to generate a figure.

Select colormap (dropdown selection) Select a colormap for your figure, if you've opted to color by normalized track time.

Azimuth viewing angles (text input) Azimuth angles over which to turn as the movie is generated, inputted as a two-element vector specifying the starting and ending azimuth angle.

Elevation viewing angles (text input) Elevation angles over which to turn as the movie is generated, inputted as a two-element vector specifying the starting and ending elevation angle.

Save path (text input) File location to save your .mp4 movie.